Additional file 1: Supplementary tables and figures for "HANFIS: A Hierarchical Adaptive Neuro-Fuzzy Inference System for Evidence-Based Dynamic Cybersecurity Risk Scoring in Cloud-Native Kubernetes Deployments"

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Abstract

Supplementary tables and figures moved out of the main text to shorten it. Each item keeps its original caption and values.

Table S1 Experimental preparation, acceptance criteria, and methodological purpose.

Preparation element	Acceptance criterion	Methodological purpose	Evidence (Runs 7–16)
Cluster health	Two Kubernetes nodes in Ready state	Excludes measurements obtained during node degradation or loss of schedulability	Gate passed in all ten pairs
Baseline workload P0	Two of two replicas Ready	Holds application capacity constant across monitoring conditions	Gate passed in all ten pairs
Service health	Internal HTTP response code 200	Confirms that the measurement target remains operational before and after state transitions	Gate passed in all ten pairs
Monitoring ON state	Five of five pods Running and fully Ready; Prometheus 1/1 Ready	Defines a complete observability treatment rather than a partially deployed stack	Gate passed in all ten pairs
Monitoring OFF state	Zero monitoring pods remaining	Defines a valid control condition without residual collector workloads	Gate passed in all ten pairs
Protocol freeze	Scripts, controller definitions, services, system fingerprint, and replication plan present	Prevents unrecorded procedural drift between runs	Applied to Runs 7–16; two aborted attempts logged as incidents
Integrity control	SHA-256 manifests validate without mismatch	Detects post hoc alteration or corruption of experimental artefacts	SHA-256 manifest retained for each of the ten run archives
Counterbalancing	Prespecified ON–OFF or OFF–ON order	Reduces order, warm-up, cache, and time-dependent bias	Five ON–OFF and five OFF–ON pairs

Supplementary tables

Supplementary figures

Supplementary notes

Supplementary note S1: Experimental platform and telemetry validation (original text of Section 3.3)

The experimental workflow was implemented in Ubuntu 24.04 under Windows Subsystem for Linux 2 (WSL 2), with Visual Studio Code connected directly to the WSL environment. Docker Desktop provided the container runtime and Kubernetes integration. The research workspace was maintained under a dedicated project hierarchy,

Table S2 E1 ablation: membership family × monotonicity, 5 seeds per cell.

Scenario	MF family	Monotone	Params.	RMSE (mean ± SD)	R^2 (mean)
S1 (smooth)	Gauss	no	176	0.093 ± 0.021	1.000
S1 (smooth)	Gauss	yes	176	0.411 ± 0.246	0.999
S1 (smooth)	Ruspini	no	147	0.043 ± 0.052	1.000
S1 (smooth)	Ruspini	yes	147	0.073 ± 0.066	1.000
S2 (threshold)	Gauss	no	176	0.117 ± 0.034	1.000
S2 (threshold)	Gauss	yes	176	6.067 ± 1.524	0.879
S2 (threshold)	Ruspini	no	147	0.646 ± 0.051	0.999
S2 (threshold)	Ruspini	yes	147	2.126 ± 0.906	0.983

Table S3 Post hoc sensitivity of the confirmatory comparison to the EPSS model change: full run versus weeks whose labels all pre-date 2026-06-15 (skill score, 95% group-clustered bootstrap CI).

Hypothesis	Comparator	Full run	Pre-change weeks
H1a	Single CVSS score	0.967 [0.951, 0.980]	0.973 [0.939, 0.991]
H1b	Persistence (4-week)	0.015 [−0.607, 0.246]	−0.055 [−2.731, 0.201]
H1c	Ridge, 7 inputs	0.696 [0.527, 0.816]	0.728 [0.484, 0.901]
H2	FlatANFISMatched	0.665 [0.269, 0.844]	0.913 [0.753, 0.974]
H2′	Flat ANFIS (full)	0.253	0.441

Table S4 Explanation-fidelity checks on the 25 saved models. Pairs are the positive-control images P0-IF with and without the active filter; S^+ denotes the 23 weeks in which exploitability differs between the two.

Check	Criterion	Result	Outcome
H6-0 trace vs. inference	$\lvert\Delta\rvert \le 10^{-9}$, all test rows	0	pass
H6-neg identical inputs	$\lvert\Delta\rvert \le 10^{-9}$, 34 weeks × 25 models	0	pass
H6a direction (monotone)	$\hat{y}_{\text{on}} - \hat{y}_{\text{off}} \ge -10^{-9}$ on S^+	minimum 0	pass
H6b panel structure	panel E unchanged, panel V changes	0 and 2.9×10^{-4}	pass
H6c deletion test	ratio ≥ 2 and proportion ≥ 0.90	4.11 and 0.835	fail

while shell procedures were stored as versionable `.sh` files rather than being executed only as transient terminal commands. This decision improved procedural transparency, reduced copy-and-paste errors, and allowed every experimental gate to be re-examined without reconstructing the original command sequence.

The Kubernetes testbed comprised two Ready nodes and a baseline application, denoted P0, deployed with two ready replicas in the `hanfis-research` namespace. A load-generator pod accessed P0 through an internal Kubernetes service. Service availability was verified using an HTTP request from inside the cluster; status code 200

Table S5 Weekly ranking utility (E15, post hoc): mean over weeks of the Spearman correlation between predicted score and realised label, and precision at the top 20%.

Horizon and metric	Formula	HANFIS	Persistence
4 weeks (45 weeks): Spearman	0.994	0.969	0.992
4 weeks: precision at top 20%	0.969	0.973	0.969
8 weeks (37 weeks): Spearman	0.989	0.963	0.982
8 weeks: precision at top 20%	0.957	0.957	0.941
12 weeks (29 weeks): Spearman	0.974	0.951	0.964
12 weeks: precision at top 20%	0.924	0.952	0.883

Table S6 Paired weekly differences in ranking utility (E15, post hoc); 95% week-bootstrap intervals in brackets.

Comparison	Δ Spearman	Δ precision at top 20%
4 w: formula − persistence	+0.002 [0.001, 0.004]	0.000 [−0.013, 0.013]
4 w: HANFIS − persistence	−0.023 [−0.027, −0.018]	+0.004 [−0.009, 0.018]
4 w: formula − HANFIS	+0.025 [0.021, 0.029]	−0.004 [−0.013, 0.000]
8 w: formula − persistence	+0.007 [0.004, 0.012]	+0.016 [0.000, 0.032]
8 w: HANFIS − persistence	−0.018 [−0.026, −0.011]	+0.016 [0.000, 0.032]
8 w: formula − HANFIS	+0.026 [0.020, 0.031]	0.000 [0.000, 0.000]
12 w: formula − persistence	+0.010 [0.005, 0.017]	+0.041 [0.014, 0.076]
12 w: HANFIS − persistence	−0.013 [−0.025, −0.002]	+0.069 [0.034, 0.103]
12 w: formula − HANFIS	+0.024 [0.017, 0.032]	−0.028 [−0.055, −0.007]

Table S7 Input-change strata at four weeks (E16, post hoc): rows, mean squared error (persistence / formula / HANFIS), and skill against persistence with 95% intervals.

Stratum	Rows	MSE	Skill, formula	Skill, HANFIS
Input changed	628	0.637 / 0.382 / 0.557	0.400 [0.308, 0.487]	0.125 [−0.402, 0.333]
No input change	452	0.107 / 0.106 / 0.202	0.002 [−0.069, 0.037]	−0.900 [−4.171, −0.206]

was required at each critical gate. The observability stack was deployed in a separate `monitoring` namespace and consisted of Prometheus The Prometheus Authors (2026), Prometheus Operator, kube-state-metrics, and node-exporter components. In its validated ON state, the stack contained five fully ready and running monitoring pods, including node-level exporters. In the OFF condition, the deployments and stateful workload were scaled to zero and the exporter DaemonSet was made unschedulable through a controlled node-selector patch. The original replica counts and Kubernetes object definitions were saved before the transition so that the monitoring state could be restored deterministically.

System preparation was organised as a sequence of auditable gates. The gates cover the baseline workload, cluster health, the observability deployment, the measurement procedure, protocol freezing and the generation of SHA-256 manifests; the criterion applied at each gate is listed in Table S1 of Additional file 1. The first gates of the protocol-development phase (O5–B4.10 to O5–B4.11B) are kept in the repository as scripts and describe the procedure only; since one of their gate reports could not be

Table S8 Calibration of four-week out-of-fold predictions (E21, post hoc). Mean error (prediction minus label), and slope β and intercept α of the regression of the label on the prediction, with 95% image-bootstrap intervals. Unaffected rows are those whose label window does not straddle the change of the EPSS model.

Predictor	Mean error	Slope β	Intercept α
All rows (1080)			
Persistence	−0.097 [−0.196, −0.013]	0.997 [0.986, 1.007]	0.109 [0.012, 0.228]
Formula	−0.040 [−0.106, 0.017]	0.998 [0.990, 1.005]	0.048 [−0.021, 0.132]
HANFIS	0.016 [−0.080, 0.104]	1.025 [1.008, 1.044]	−0.115 [−0.228, 0.007]
Unaffected rows (890)			
Persistence	−0.015 [−0.079, 0.025]	0.997 [0.993, 1.001]	0.026 [−0.017, 0.091]
Formula	0.012 [−0.029, 0.040]	1.000 [0.997, 1.002]	−0.012 [−0.044, 0.035]
HANFIS	0.068 [−0.029, 0.157]	1.031 [1.014, 1.053]	−0.193 [−0.300, −0.085]

reconstructed after the fact, they are not cited as evidence of passing. The evidence that the gates were satisfied is the set of run archives and manifests for Runs 7–16 (Section 3.4), in each of which both conditions were admitted only after the gate criteria held. Because P0 stayed at two of two ready replicas with HTTP status 200 across every admitted transition, the change in observability state did not itself interrupt the target service.

Supplementary note S2: Ablation and robustness analysis (original text of Section 4.5)

Table S2 of Additional file 1 reports the completed membership-family×monotonicity ablation (E1) on two synthetic scenarios designed to stress-test the training scheme before it is committed to real data. S1 is a smooth response with no thresholds and S2 a response containing a hard threshold; each is fit with five seeds at the 176-parameter (Gaussian) or 147-parameter (Ruspini) configuration.

The monotonicity constraint costs little on the smooth scenario but is expensive on the threshold scenario for the Gaussian family specifically (RMSE 0.117 → 6.067). A hard threshold is locally non-monotone, and softplus-constrained slopes cannot represent the local reversal. The connected Ruspini family degrades far less under the same constraint (0.646 → 2.126), because its partition of unity concentrates the approximation error near the threshold instead of distributing a large penalty across every rule. Because the real target in Section 3.5 is not known in advance to be smooth or threshold-like, amendment A-10 retains the Gaussian family with monotonicity active as the frozen default (consistent with the higher-is-riskier requirement) and treats Ruspini as a documented, cheaper fallback if the monotone-Gaussian cost proves large on real folds. An end-to-end pure-Adam variant without the closed-form anchor (Section 3.2) was also tested during this ablation and diverged on a subset of seeds under otherwise identical settings, confirming that the hybrid scheme is necessary.

Sensitivity to the EPSS model change. The EPSS scoring model changed version on 2026-06-15 (`v2025.03.14` to `v2026.06.15`). Because the label of week t is computed from snapshots $t{+}7, \ldots, t{+}28$ days ahead, two of the five held-out blocks of the confirmatory comparison involve labels computed under the newer version: one block straddles the change and one lies wholly after it. This contrast was not pre-registered, so it is reported as a post hoc sensitivity analysis. The confirmatory pipeline was



Fig. S1 Methodological mind map linking the cloud-native testbed, telemetry acquisition, monitoring-overhead controls, reproducibility evidence, and HANFIS integration.

re-run with the same code, seeds and frozen configuration on the 4,560 asset-weeks whose labels all pre-date the change (weeks up to 2026-05-11; 720 held-out rows), with the results in Table S3 of Additional file 1. H1a, H1c and H2 remain confirmed, and H1b remains unconfirmed, with a much wider interval. Because both the training and the held-out windows change, skill scores are not directly comparable across the two columns. The larger margin over the matched flat model (H2: 0.665 to 0.913) comes from the matched model, whose mean squared error per held-out row rises from 1.22 to 3.77 while that of HANFIS stays at 0.41 and 0.33.

Where HANFIS and persistence differ. The pooled skill against persistence (0.015) hides two regimes, and this exploratory, post hoc analysis (not part of the Holm-corrected family) separates them. Dropping one held-out fold at a time leaves it

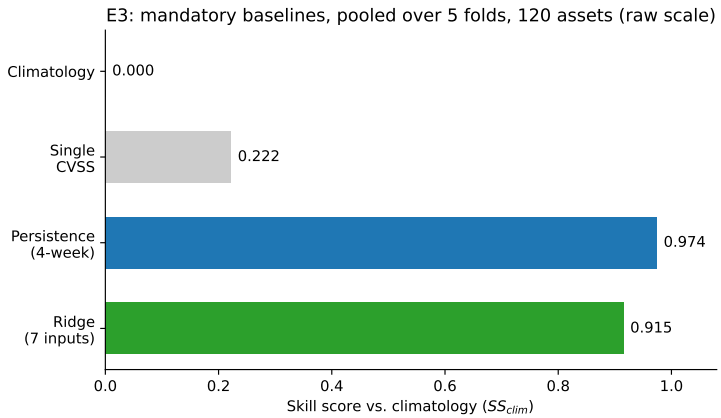


Fig. S2 Mandatory baselines' skill against climatology, pooled over 5 folds. A single CVSS score now adds modest skill over climatology (22%, up from essentially nothing on the earlier 27-image pilot); persistence and ridge separately recover over 90% of the climatology-relative skill each, though not of each other; see the mandatory-baselines table in the main text for the reversal in their relative standing.

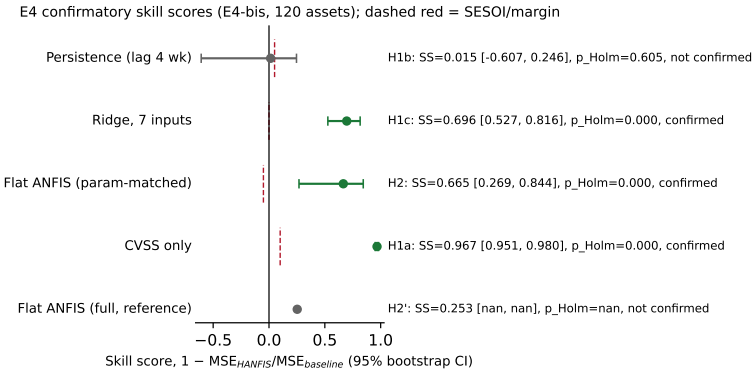


Fig. S3 E4 confirmatory results: HANFIS skill score against each comparator with 95% group-clustered bootstrap confidence intervals. Green bars exclude zero (confirmed); the red bar (H1b, vs. persistence) does not.

between 0.007 and 0.172 except for fold 3, whose removal gives −1.168. Fold 3 ends at the EPSS model change, and the label of week t is built from snapshots up to t+28 days ahead whereas the persistence forecast reuses the label of week t−4. For the four weeks in which the label is fully computed under the new model while the lagged label is still computed under the old one (2026-W24 to W27; 95 rows), skill against persistence is +0.483; on the remaining 985 rows it is −0.420. In 2026-W25 and W26 the squared error of persistence is 63.1 and 28.9 against 0.7 and 1.0 for HANFIS, while the parameter-matched model also adapts (4.8 and 5.2), so the advantage reflects access to the current EPSS input rather than the hierarchical structure. In the week the label first jumps (W24) neither HANFIS nor persistence anticipates it (108.9 and 115.2). In

the stable weeks that follow HANFIS keeps a small error floor (about 0.7–0.8 per week over 24 rows, against 0.1 for persistence). The pooled advantage of 6.5 squared-error units is thus the sum of +104.1 in the break weeks and −97.6 elsewhere. The same pattern appears before the change: in the run restricted to earlier weeks the advantage concentrates in 2026-W06 and W07, the weeks of one image's label surge (an image carrying 77.8% of the persistence error), and the total is −12.3. HANFIS is therefore reactive rather than predictive: better than persistence when the label moves, worse when it does not. Two further observations qualify H2. The first held-out fold, trained on 2.2 observations per parameter, contributes 91% of the pooled gap to the matched model (squared error 869.6 against 73.9). Without it the skill against FlatANFIS-Matched is 0.183 and against the full flat model 0.084, whereas the skill against the ridge model stays between 0.656 and 0.793 whichever fold is removed.

Supplementary note S3: Explanation fidelity (original text of Section 4.6)

We tested whether the rule-level explanation that HANFIS exposes is faithful to its inference. The test uses the 25 saved models (five folds × five seeds), which reproduce the confirmatory predictions of Section 4 exactly (mean over the five seeds; maximum difference 1.4×10^{-6} on the 0–100 scale in single precision). The hypotheses and thresholds were drafted and announced before execution; the version-control freeze, however, preceded the commit that contained the protocol text, so the protocol is not frozen in the strict sense and the run is reported as a documented deviation. Table S4 of Additional file 1 lists the outcome.

Removing the three rules ranked highest by attribution changed the output 4.1 times more than removing three random rules, but it exceeded the median random removal in only 83.5% of non-degenerate rows (criterion: 90%); 456 of 5,400 rows (8.4%) were degenerate, 446 of them because the output was clamped. Of the 575 positive-control pairs, 552 increased strictly; the remaining 23 include 20 pairs whose output is clamped at zero in both conditions. Post hoc and without bearing on the decision, the proportion reached 0.90 in two of five folds and was lowest in fold 3 (0.637), and it was lowest on the P0 and P0-IF rows (0.648, ratio 1.73, 135 rows). On the nine paired test weeks the mean ON–OFF contrast was 0.037 for the model and 0.073 for the label (0–100 scale): the direction is right and the magnitude is about half.

We therefore do not claim validated rule-level explanations. The rule trace matches the inference exactly and the direction of the response is verified, but the attribution of the output to individual rules is faithful on average rather than uniformly.

Supplementary note S4: Prioritisation utility (original text of Section 4.10)

Risk scores are used to rank assets, so we also compared the predictors on ranking within each week rather than on squared error. For every week with at least 15 assets we computed the Spearman correlation between the predicted score and the realised label and the precision at the top 20% (the share of the 20% of assets with the highest predicted score that are also among the 20% with the highest realised label), and

averaged over weeks. Differences between predictors are paired by week, with 95% intervals from a bootstrap that resamples weeks, a more conservative unit than images. This analysis is post hoc and uses the out-of-fold predictions at the three horizons.

All predictors rank assets almost identically (Table S5 of Additional file 1). At four weeks the weekly Spearman correlation is 0.99 for the formula and for persistence and 0.97 for HANFIS, and the precision at the top 20% is 0.97 for all three, so the lower squared error of the formula does not translate into a practical gain in prioritisation (Table S6 of Additional file 1: Spearman +0.002 and precision 0.000 against persistence). At eight and twelve weeks persistence ranks somewhat worse (precision 0.94 and 0.88) and both HANFIS and the formula improve on it in precision (+0.016 and +0.041 for the formula, +0.016 and +0.069 for HANFIS). At twelve weeks HANFIS identifies the top 20% better than the formula (+0.028, interval [0.007, 0.055]) although its Spearman correlation is lower (−0.024). Differences in Spearman correlation never exceed 0.03 and the precision is close to its ceiling, so all these differences should be read as small; with eighteen paired comparisons and no correction for multiplicity, the intervals are descriptive.

Supplementary note S5: Error anatomy and the EPSS model change (original text of Section 4.11)

The preceding sections compare pooled errors. To see where the differences come from, we analysed the four-week out-of-fold predictions by stratum, by time block and by distance to the change of the EPSS model on 2026-06-15 (post hoc, with a bootstrap clustered by image).

Reactivity. The advantage of the formula over persistence is concentrated in rows in which an input changed during the previous four weeks (EPSS exploitability or KEV status; 628 of 1,080 rows). Its skill is 0.400 (interval 0.308 to 0.487) in these rows and 0.002 (−0.069 to 0.037) where nothing changed, so it behaves like persistence when nothing happens (Table S7 of Additional file 1); a gate that returns persistence in unchanged rows leaves it unchanged. HANFIS has skill 0.125 in changed rows and −0.900 (−4.171 to −0.206) in unchanged rows: it adds noise where persistence is already adequate. A gate removes this deficit (improvement 0.098, interval 0.033 to 0.204) but leaves HANFIS no better than persistence (0.111, −0.367 to 0.292) and worse than the formula (−0.383, −0.998 to −0.131). Since the flag counts any change of the score, most flagged changes are small.

Squared error is extremely concentrated: the worst 5% of rows hold 91% (persistence), 95% (formula) and 67% (HANFIS) of the total, and the five worst images 68%, 63% and 39%. Leaving out each of the five time blocks in turn leaves the skill of the formula against persistence between 0.34 and 0.43, with intervals above zero, although one block (April to June 2026) carries 73% of the absolute reduction in error; HANFIS is worse than the formula in every such estimate.

The model change. The largest errors are not KEV events (the KEV indicator is zero in all twelve worst rows of every predictor) but small shifts of the score, with labels between 3.7 and 10 points on a 100-point scale, in the weeks 2026-W21 to W24. The four-week label windows of these weeks straddle the change of the EPSS model: x_2

is computed with the old model while the label, and the label of the persistence forecast, use the new one, and all three predictors err almost equally there. Rows within the label window of the change (190 of 1,080 at four weeks) hold 77% of the squared error of persistence and of the formula and 49% of that of HANFIS. On the remaining rows (Table 9) the formula keeps its point estimate of skill against persistence (0.348 against 0.357) but the interval now includes zero (−0.060 to 0.422). HANFIS is significantly worse than persistence (−1.228) and than the formula (−2.418). The forward evaluation described above is free of this change: its test weeks start on 2026-08-31 and its persistence labels come from weeks after 2026-08-03.

Horizon, revisited. The narrowing of the gap between HANFIS and the formula at eight and twelve weeks (Section 4.9) must be read in this light. Excluding the affected rows, HANFIS stays behind the formula at eight and twelve weeks (−0.324 and −0.362, intervals including zero). On the 169 asset-weeks that are common to the three horizons and unaffected at all of them, the skill of HANFIS against the formula still rises with the horizon (−15.4, −1.1 and −0.5; change between four and twelve weeks +14.9, interval 6.1 to 53.7). It stays significantly negative at every horizon (−0.515, interval −2.117 to −0.092, at twelve weeks). The horizon sweep therefore shows a shrinking deficit, not parity; the sample of unaffected rows is small and the intervals are wide.

Supplementary note S6: Calibration, recalibration and stacking, E21 to E24 (original text of Section 4.12)

We checked calibration after the registration of the forward evaluation had become public (post hoc, descriptive, no training). For the four-week out-of-fold predictions of persistence, the formula and HANFIS we report the mean error (prediction minus label), the slope β and intercept α of the regression of the label on the prediction (a calibrated predictor has $\beta = 1$ and $\alpha = 0$), and the mean label by decile of the prediction. Intervals come from the bootstrap clustered by image used above. The coefficients of the formula were fitted inside each training fold, on the same folds as HANFIS, so its calibration is also out-of-fold. Table S8 of Additional file 1 covers all 1,080 rows and the 890 rows whose label window does not straddle the change of the EPSS model (Section 4.11).

All three predictors lie close to the identity line. Persistence runs low on all rows (mean error −0.097, interval −0.196 to −0.013), and almost all of this bias comes from the weeks around the model change: on unaffected rows it is −0.015 (−0.079 to 0.025). The interval of the formula's mean error includes zero in both samples. HANFIS departs most: its slope is 1.025 (1.008 to 1.044) on all rows and 1.031 (1.014 to 1.053) on unaffected rows, where its intercept is −0.193 (−0.300 to −0.085). In the fourth decile of its predictions the mean prediction is 1.0 and the mean label 0.5 (unaffected rows: 1.1 and 0.5); the mean label does not rise between the third and fourth decile (0.5 in both) although the mean prediction doubles. In the upper deciles predictions and labels agree (7.9 and 8.1 in the eighth decile, 9.8 and 9.9 in the ninth). These departures are small next to the label scale. The intervals above resample images and treat the weekly shocks shared by all images as fixed. A bootstrap clustered by week (E22, post hoc; 45 weeks, 37 of them unaffected by the model change) gives narrower

intervals for these statistics, not wider ones: for HANFIS the slope is 1.025 (1.015 to 1.035) on all rows and 1.031 (1.020 to 1.042) on unaffected rows, and the intercept −0.115 (−0.189 to −0.037) and −0.193 (−0.242 to −0.147). To compare predictors we took the distance from the ideal value ($\beta = 1$, $\alpha = 0$, mean error 0) and subtracted that of the formula from that of HANFIS, so that a positive number means HANFIS is further away. For the slope the excess is 0.023 (0.011 to 0.033) on all rows and 0.031 (0.018 to 0.042) on unaffected rows; with intervals clustered by image the first is 0.023 (0.001 to 0.041), a borderline result. On all rows the excess in the intercept and in the mean error includes zero (0.068, −0.072 to 0.179, and −0.024, −0.061 to 0.061; week-clustered). On unaffected rows the excess in the slope and in the intercept is positive under both kinds of clustering (slope 0.031: week 0.018 to 0.042, image 0.013 to 0.051; intercept 0.182: week 0.131 to 0.230, image 0.063 to 0.278). The excess in the mean error (0.055), by contrast, is positive only with the week-clustered interval (0.034 to 0.068; image: −0.016 to 0.133). Against persistence the pattern on unaffected rows is the same: slope 0.028 (week 0.015 to 0.040, image 0.011 to 0.050) and intercept 0.167 (week 0.111 to 0.219, image 0.021 to 0.286) are positive under both, the mean error 0.053 (week 0.005 to 0.088, image −0.053 to 0.146) only with week clustering. We read the miscalibration of HANFIS as small but consistent in slope and intercept where the model change does not interfere, and as descriptive.

Recalibration (E23). We asked whether a linear correction could close the gap between HANFIS and the formula. For this we refitted label = $a + b\times$ prediction for each predictor, once on the rows of the other folds (cross-fitted) and once on all rows (in-sample, an optimistic ceiling), and recomputed the skill of HANFIS against the formula (post hoc; intervals clustered by image). Cross-fitting uses rows from later folds, so this is a diagnostic and not a forecasting procedure. On all rows the skill is −0.534 (−1.383 to −0.204) before, −0.535 (−1.287 to −0.232) after the cross-fitted and −0.509 (−1.223 to −0.213) after the in-sample recalibration; the change from the cross-fitted recalibration is −0.001 (−0.047 to 0.116). On unaffected rows the values are −2.418 (−13.204 to −0.709), −2.315 (−12.609 to −0.702) and −2.175 (−12.388 to −0.647); the change of 0.103 (−0.152 to 1.060) includes zero and is about 4% of the deficit (10% for the in-sample ceiling). The deficit of HANFIS is therefore not explained by a linear miscalibration: it remains after the correction, in line with the noise HANFIS adds in rows where no input changed (Table S7 of Additional file 1).

Stacking (E24). We asked whether HANFIS carries information that the formula lacks. For this we fitted label = $a + b_1\times$ formula $+ b_2\times$ HANFIS on the rows of the other folds, applied it to each fold in turn (cross-fitted), and compared it with the formula recalibrated in the same way (post hoc; intervals clustered by image). On all rows the skill of the combination against the recalibrated formula is −0.001 (−0.006 to 0.004); the best case, with coefficients fitted on all rows, is 0.0005 (−0.002 to 0.003), and the weight of HANFIS is 0.030 (between folds 0.015 to 0.052). Persistence as the second predictor instead of HANFIS gives 0.001 (−0.022 to 0.021). On unaffected rows the values are −0.002 (−0.028 to 0.017) for HANFIS and −0.169 (−0.224 to −0.005) for persistence; the second reflects unstable coefficients, since the weights of formula and persistence vary widely between folds when both are included. A linear

combination of the two predictors therefore does not beat the formula: HANFIS adds no information that a straight line can use.

Supplementary note S7: Limitations (original text of Section 5, subsection Limitations)

Two limits of construct validity come first. First, the label and two of the inputs share ingredients: x_2 (EPSS $\times\,\rho$) and x_7 (the KEV indicator) are the values at t of quantities that the label aggregates over $t{+}7, \ldots, t{+}28$, and runtime reachability ρ enters both. The task is therefore to forecast the near future of exploitability signals that the model can already see, not to predict independent compromise outcomes, for which no ground truth exists here. The persistence baseline, built in the same way, is the relevant bar, and the comparison with a single CVSS score should be read in that light. The exact Shapley attribution (Section 4.7) and the ablation without these two inputs (Section 4.8) show how heavily the model's skill rests on them. Second, the dynamic character of the score comes from the time-varying EPSS and KEV feeds and from runtime reachability. Runtime telemetry (CPU, memory, latency) is used only to quantify observability overhead and is not a model input, so "dynamic" must not be read as telemetry-driven scoring. The rule-level attribution is faithful on average but not uniformly (Section 4.6), and we do not claim validated explanations.

All analyses labelled post hoc were designed after the confirmatory results and run on the same data; they are descriptive, carry no multiplicity correction, and none of them changes a registered decision.

The inventory consists of public container images and a small number of testbed assets built from a public base image, and the label reflects exploitation signals of public vulnerabilities (EPSS, KEV), not incidents in a specific organisation; generalisation to other workloads or to incident-based outcomes is untested.

The confirmatory comparison (Table 4) uses the full 120-asset inventory. On the same inventory (6,357 labelled rows; Table 3), a linear ridge model over all seven inputs underperforms persistence on every one of the five grouped-temporal folds ($\mathrm{SS}_{\mathrm{pers}} = -2.293$ pooled, raw scale). We suspect that the linear model over-extrapolates as the inventory's heterogeneity grows, where persistence's week-to-week copy remains comparatively safe, but we did not test this explanation.

The EPSS scoring model changed version (`v2025.03.14` to `v2026.06.15`) inside the sealed 57-week window, a threat to label comparability across that boundary. Two of the five held-out blocks involve labels computed under the newer version, and a post hoc re-analysis restricted to weeks whose labels all pre-date the change leaves the confirmatory conclusions unchanged (Section 4.5).

The data-sufficiency gate (at least ten observations per parameter) is met by the pooled sample (6,357 labelled rows against 176 parameters) but not by the two earliest held-out folds (2.2 and 7.1 observations per parameter), which are also those in which HANFIS is furthest behind persistence. H2's pooled margin is dominated by the first fold (Section 4.5).

CVEs lacking a Trivy-reported publication date are dated by their first appearance in a sealed EPSS snapshot, which is conservative (at most 7 days late) but not exact. A single CVE whose Trivy status lies outside the accepted set (for example "under

investigation" for a freshly published Red Hat advisory) makes the whole asset-week unusable. This affected one of the 120 images for seven consecutive weeks from 2026-08-10, and it can reach at most the two Red Hat based images of the inventory. Such rows are removed from the analysis.

Labels whose four-week window straddles the EPSS model change of 2026-06-15 (and the persistence labels derived from them) mix two score definitions; these rows (18% of the out-of-fold rows) hold 77% of the squared error of persistence and of the formula, and the forward evaluation avoids them.

The label aggregation rule (a row-wise maximum over active CVEs), its numeric scale, and the operational definition of Patch Risk were closed and pre-registered as amendment A-13 (cross-checked against the literal target-definition formula, Section 3.5) before the confirmatory comparison in Table 4 was run.

The testbed is a single two-node WSL 2/Docker Desktop Kubernetes cluster; external validity to production multi-node clusters and other CNI/storage stacks is untested. Ten paired observability runs, while showing large and consistent effects, still provide limited power for null-hypothesis testing on the latency measure specifically, which is why the reported confidence intervals rather than a significance threshold carry the interpretive weight in Section 4.1.

References

The Prometheus Authors (2026) Prometheus: Overview. https://prometheus.io/docs/introduction/overview/, accessed 14 October 2026